

ENVS-193DS Final

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[Link to repo](#)

```
# read in packages
library(tidyverse)
library(here)
library(DHARMA)

# reading in data and creating new object
nest <- read_csv(
  here("data", "nest_data_final.csv"))
```

Problem 1. Research writing

Problem 2. Data Analysis

a.

```
# creating clean data frame
nests_clean <- nest |>
  # selecting desired columns
  select(ant_species, height_cm, case_control) |>
  # filtering for desired ant species
  filter(ant_species %in% c("RRB", "BBR", "AB")) |>
  # creating new column for full scientific name
  mutate(sci_name = case_when(
    ant_species == "RRB" ~ "Crematogaster sjostedti",
    ant_species == "BBR" ~ "Crematogaster mimosae",
    ant_species == "AB" ~ "Crematogaster nigriceps"),
    # reordering factors for ant species
    ant_species = factor(ant_species,
      levels = c("RRB", "BBR", "AB"))
  )

# displaying structure of data frame
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species : Factor w/ 3 levels "RRB","BBR","AB": 1 1 1 1 3 1 2 1 3 2 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ sci_name    : chr [1:270] "Crematogaster sjostedti" "Crematogaster sjostedti" "Crematogaster sjostedti" ...
```

```
# slicing sample
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
  ant_species height_cm case_control sci_name
  <fct>         <dbl>         <dbl> <chr>
1 RRB           392.             1 Crematogaster sjostedti
2 RRB           121             0 Crematogaster sjostedti
```

3	RRB	164.	0	Crematogaster sjostedti
4	AB	85	0	Crematogaster nigriceps
5	BBR	155	0	Crematogaster mimosae
6	RRB	215	0	Crematogaster sjostedti
7	AB	132.	0	Crematogaster nigriceps
8	BBR	141	0	Crematogaster mimosae
9	RRB	119	0	Crematogaster sjostedti
10	RRB	225	0	Crematogaster sjostedti

b.

For this data, a value of 1 in the case_control column indicates that a nest was observed in the acacia tree in question. A value of zero in that said column means that no bird nest was observed in the tree.

c.

Tree Height (height_cm)

Tree height in this data set is a continuous variable that is measured in centimeters. As tree height increases the probability of nest occurrence would likely increase. Taller trees are more structurally complex and therefore have more nesting opportunities, they may also provide better predator avoidance for birds (Introduction, paragraph 2; Methods, “Study site and species” subsection, paragraph 1).

Acacia ant species (ant_species)

As ant species changes, the probability of nest occurrence would likely differ. This is due to each ant species exhibiting a distinct aggression level and mutualistic relationship with the host Acacia tree which affects the level of predator protection a nest in said tree would receive. (Introduction, paragraph 3; Methods, “Nest box experiment” subsection, paragraph 1)

d.

Model number	Ant Species	Tree Height	Predictor list
0	X	X	all predictors, no interaction (null model)
1	X	X	all predictors, with interaction (full model)